

Notes: Gruber (AER, 1994)

The Incidence of Mandated Maternity Benefits

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1 Big picture

- **Question.** When government mandates employers to provide a benefit to a demographically identifiable group, who pays for the mandate: firms through higher labor costs, workers through lower wages, or both through reduced employment and hours?
- **Policy setting.** The paper studies state and federal laws in the 1970s that required health insurance plans to cover childbirth comprehensively, rather than excluding maternity or paying only limited flat benefits.
- **Targeted group.** The mandates raised the relative cost of employing women of childbearing age and, through family insurance coverage, potentially their husbands as well.
- **Central economic issue.** Mandated benefits can be efficient if workers value the benefit and wages adjust downward by the cost of the benefit. For group-specific mandates, wage adjustment may be blocked by antidiscrimination law, internal pay norms, union rules, or fairness constraints.
- **Empirical strategy.** The paper uses state maternity mandates before the federal Pregnancy Discrimination Act (PDA) and then uses the federal PDA as a reverse experiment. It compares targeted groups to control groups, experimental states to nonexperimental states, and before-law to after-law periods.
- **Main finding.** The cost of the maternity-benefit mandates was substantially shifted to wages of the targeted groups, with estimates often close to full shifting.
- **Labor-input finding.** There is little evidence of a large net decline in employment, hours, or total labor input for the targeted groups.
- **Policy implication.** Group-specific mandates need not raise employers' effective cost of employing the target group if wages can adjust. This makes mandates potentially efficient as benefit financing instruments, though not necessarily distributionally desirable.
- **Broader contribution.** The paper is an early and influential empirical example of using natural experiments and difference-in-differences logic to study the incidence of workplace benefits.

2 Institutional background and economic mechanism

2.1 Health insurance treatment of maternity before 1978

- Before the state and federal reforms, maternity coverage in employer health insurance was often worse than coverage for comparable illnesses.
- Differential treatment took two main forms: no pregnancy coverage, or flat cash benefits that did not cover actual childbirth costs.
- Gruber uses the 1977 National Medical Care Expenditure Survey to estimate the extent of differential coverage. Among women aged 20–40 with employment-based group health insurance, about 20 percent lacked maternity coverage despite coverage for comparable inpatient services.
- Another roughly 30 percent received less generous reimbursement for physicians' maternity services or only a low flat payment for delivery.
- Therefore, at least half of covered women of childbearing age faced differential maternity coverage or benefits before the mandate environment became universal.

Interpretation: The mandates mattered because they changed real insurance coverage, not merely legal language. They forced employers or insurers to equalize maternity coverage with other medical conditions.

2.2 State mandates and the federal Pregnancy Discrimination Act

- During 1975–1979, many states enacted laws requiring pregnancy to be treated like comparable illnesses in health insurance benefits.
- The key state-law experiment focuses on states that adopted mandates in the mid-1970s before the federal law.
- In 1978, the federal Pregnancy Discrimination Act prohibited differential treatment of pregnancy in the employment relationship.
- The federal law is used as a second experiment because states that had already passed maternity mandates were less affected by the federal expansion, while states without prior laws became newly treated.

Interpretation: The state and federal laws create two related but distinct sources of policy variation: first, early adopting states versus nonadopting states; second, the federal law applied to previously untreated states while previously treated states serve as controls.

2.3 Why group-specific mandates are theoretically ambiguous

The standard mandate-incidence logic is:

$$\text{Mandated benefit cost} \Rightarrow \text{lower wages if workers value the benefit.} \quad (1)$$

If employees value the benefit at its cost, wages can fall by the full cost:

$$\Delta W = -C, \quad (2)$$

where C is the employer cost of the benefit.

- If wages fully adjust, the mandate acts like a benefits tax paid by workers.
- If wages cannot adjust, employers bear the cost and may reduce employment or hours of the targeted group.
- Group-specific mandates are more difficult than general mandates because they require relative wages to adjust across groups inside firms.

Interpretation: The empirical question is whether relative wages actually fall for the targeted demographic group after the mandate. If they do, the mandate is shifted to workers; if they do not, employers face a higher cost of employing the targeted group.

2.4 Cost of expanding maternity benefits

- The paper estimates the cost of adding maternity benefits using a premium-calculation package from a national insurer for small-group health plans.
- Costs are calculated separately by demographic group and coverage type.
- Young women with family coverage face the highest estimated cost because the probability of childbirth is highest for this group.
- Table 1 reports annual costs in 1990 dollars, annual costs in 1978 dollars, and the cost as a percentage of 1978 weekly earnings.

Interpretation: These costs provide the scale needed for the wage-incidence tests. They also show that the mandates were large enough to produce measurable wage effects.

3 Data and measurement

3.1 Core data source: Current Population Survey

- The main labor-market outcomes come from the Current Population Survey (CPS).
- The paper uses CPS data to measure hourly wages, weekly hours, employment, and demographic characteristics.
- The main analysis focuses on wage earners and compares targeted workers with control workers.
- The paper uses May CPS data for the state-law experiment and March CPS data for the federal experiment.

Interpretation: The CPS allows the paper to identify demographic groups affected by maternity-benefit mandates and to observe labor-market outcomes before and after legal changes.

3.2 Treatment groups

The mandates directly or indirectly affect three demographic groups:

1. married women aged 20–40;
2. single women aged 20–40;
3. married men aged 20–40, because their family coverage may include spouses’ maternity benefits.

The main treatment group in the simplest state-law analysis is married women aged 20–40.

Interpretation: The target group is defined by demographic exposure to maternity costs. Married women are the cleanest group because both their own fertility risk and their household insurance coverage are relevant.

3.3 Control groups

The benchmark control group consists of:

- men and women older than 40; and
- single men aged 20–40.

These workers should be less affected by maternity-benefit mandates.

Interpretation: The control group is intended to capture labor-market trends in the same states and time periods that are unrelated to the mandate’s group-specific cost shock.

3.4 Experimental and nonexperimental states

- Experimental states are states that passed maternity mandates in the relevant state-law period and are separately identifiable in the CPS.
- Nonexperimental states are comparable states that did not pass such laws before the federal PDA.
- For the federal experiment, the logic reverses: states that had already passed mandates form a control group for states newly affected by the federal law.

Interpretation: The state variation supplies the quasi-experimental treatment intensity needed to distinguish mandate effects from national trends affecting all workers.

3.5 Outcome variables

The main outcomes are:

- log hourly wage;
- log weekly hours;
- employment probability;
- total labor input, combining employment and hours.

Interpretation: Wage changes identify incidence. Hours and employment changes identify whether the mandate changes the quantity of labor demanded or supplied.

4 Empirical specifications and identification

4.1 Triple-difference logic

The simplest design is a difference-in-difference-in-differences (DDD). Let:

- $Treat_i = 1$ for the targeted demographic group;
- $Exp_s = 1$ for states passing the maternity mandate;
- $Post_t = 1$ for the post-law period.

A generic DDD regression is:

$$Y_{ist} = \alpha + \beta_1 Treat_i + \beta_2 Exp_s + \beta_3 Post_t + \beta_4 (Treat_i \times Exp_s) \quad (3)$$

$$+ \beta_5 (Treat_i \times Post_t) + \beta_6 (Exp_s \times Post_t) + \delta (Treat_i \times Exp_s \times Post_t) + X'_{ist} \gamma + \varepsilon_{ist}. \quad (4)$$

- Y_{ist} is log wage, log hours, employment, or labor input.
- X_{ist} includes demographic controls.
- δ is the policy effect on the targeted group in mandate states after the law, net of group trends, state trends, and period effects.

Interpretation: The DDD coefficient removes three sources of confounding: permanent differences between treatment and control groups, permanent differences between states, and common time shocks.

4.2 Parametrized cost specification

The treatment dummy design treats all affected workers as equally exposed. The paper also estimates exposure using individual-level predicted mandate costs.

A generic parametrized equation is:

$$Y_{ist} = \alpha + \lambda Cost_{ist} + X'_{ist} \gamma + \eta_s + \tau_t + \varepsilon_{ist}, \quad (5)$$

where $Cost_{ist}$ is the predicted cost of the maternity mandate for worker i .

The cost is constructed from:

- the age-specific premium cost of maternity coverage;
- whether the worker is likely to have employer health insurance;
- whether coverage is individual or family coverage;
- the relevant demographic category.

Interpretation: The cost specification uses within-treatment variation: workers expected to cost more under the mandate should experience larger wage reductions if the mandate is shifted to wages.

4.3 Shifting percentage

The paper converts coefficients into a shifting percentage:

$$\text{Shifting percentage} \approx 100 \times \frac{\text{estimated wage reduction}}{\text{estimated mandate cost}}. \quad (6)$$

- A shifting estimate of 100 percent means the wage falls by the cost of the mandate.
- A shifting estimate below 100 percent means employers or other parties bear part of the cost.
- A shifting estimate above 100 percent can arise from sampling error, imperfect cost measurement, or over-adjustment.

Interpretation: The empirical question is not merely whether wages fall, but whether the magnitude of the wage fall matches the actuarial cost of the mandated benefit.

4.4 Federal reverse experiment

The federal PDA creates a second design:

- states that had already passed maternity mandates should be less affected by the federal law;
- states without prior mandates become newly treated by the federal law.

Interpretation: This reverse experiment is a strong robustness check. If the state-law results are truly due to maternity mandates, the federal law should generate similar wage-shifting patterns in states that were newly affected.

4.5 Identifying assumptions

The main assumptions are:

- absent the mandate, targeted workers in mandate states would have followed the same relative wage trends as comparable workers in nonmandate states;
- no other policy or shock differentially affected women of childbearing age in mandate states at the same time;
- the control groups are not themselves substantially affected by the mandate;
- measured mandate costs are a reasonable proxy for true employer costs.

Interpretation: These assumptions are why the paper uses both state mandates and the federal reverse experiment, and why it checks multiple demographic groups and labor-input outcomes.

5 Tables

5.1 Table 1: Cost of adding maternity benefits

Read:

- Table 1 reports the cost of adding maternity benefits to a health insurance package for six demographic and coverage groups.
- Family coverage for women aged 20–29 is the most expensive category: \$984 in 1990 dollars, or \$360 in 1978 dollars.
- Family coverage for women aged 30–39 costs \$756 in 1990 dollars, or \$277 in 1978 dollars.
- Individual coverage is much cheaper than family coverage because family coverage includes maternity costs for dependents.
- Expressed as a percentage of 1978 weekly earnings, the cost is large enough to produce detectable wage changes.

Economic message: The table gives the dollar scale of the mandate. Without these cost estimates, it would be impossible to judge whether wage declines represent full or partial shifting.

Coverage	Demographic group	Annual cost (1990 dollars)	Annual cost (1978 dollars)	Cost as percentage of 1978 weekly earnings
Family	20–29-year-old females	\$984	\$360	4.6
Family	30–39-year-old females	\$756	\$277	3.5
Individual	20–29-year-old females	\$324	\$119	1.5
Individual	30–39-year-old females	\$252	\$92	0.9
Family	20–29-year-old males	\$984	\$360	2.9
Family	30–39-year-old males	\$756	\$277	1.7

Notes: The source of the data is a premium-calculation program from an anonymous insurance carrier. The cost was calculated for a two-person firm in Maryland. Maryland was a location which was approximately at the midpoint of the locational cost distribution. The results are not sensitive to variations in firm size. Costs are for 1990; they are deflated to 1978 using a weighted average of the detailed CPI for hospital services and physician services, where the weights are $\frac{2}{3}$ and $\frac{1}{3}$, respectively. Costs are normalized by 1978 weekly wages from the May 1978 CPS. For single coverage, wages of unmarried persons are used; for family coverage, wages of married persons are used.

Variable	State laws				Federal PDA			
	Nonexperimental states		Experimental states		Nonexperimental states		Experimental states	
	Before law	After law	Before law	After law	Before law	After law	Before law	After law
Percentage female	41.4 [49.3]	43.9 [49.6]	41.4 [49.3]	43.1 [49.5]	44.4 [49.7]	45.6 [49.8]	44.5 [49.7]	45.8 [49.8]
Average age	38.1 [12.6]	37.6 [12.5]	38.9 [12.6]	38.4 [12.6]	37.6 [12.7]	37.5 [12.4]	37.5 [12.7]	37.2 [12.4]
Percentage married	75.0 [43.3]	70.8 [45.5]	71.6 [45.1]	67.9 [46.7]	65.7 [47.5]	63.8 [48.1]	70.0 [45.8]	67.0 [47.0]
Percentage nonwhite	8.8 [28.3]	9.2 [28.9]	10.2 [30.3]	12.0 [32.5]	12.3 [32.8]	13.5 [34.2]	10.9 [31.2]	11.2 [31.4]
Average education	12.1 [2.87]	12.3 [2.81]	12.4 [2.94]	12.7 [2.88]	12.5 [3.04]	12.7 [2.99]	12.3 [2.97]	12.6 [2.88]
Average hourly wage	5.68 [3.31]	5.59 [3.16]	6.61 [3.98]	6.40 [3.62]	6.33 [4.02]	5.88 [3.74]	5.80 [3.81]	5.49 [3.63]
Percentage union	27.0 [44.4]	26.8 [44.3]	33.4 [47.2]	33.8 [47.3]				
Percentage manufacturing	36.5 [48.2]	35.3 [47.8]	28.5 [45.1]	26.6 [44.2]	25.1 [43.3]	23.9 [42.6]	23.6 [42.5]	22.6 [41.8]
Percentage services	29.7 [47.2]	31.5 [46.5]	35.3 [47.8]	37.3 [48.4]	41.4 [49.3]	42.6 [49.4]	39.9 [49.0]	40.9 [49.2]
Weekly cost of mandate	4.01 [1.68]	3.87 [1.65]	3.92 [1.66]	3.85 [1.63]	3.64 [1.59]	3.56 [1.58]	3.71 [1.60]	3.64 [1.59]
Cost/wages	0.020 [0.010]	0.020 [0.010]	0.018 [0.009]	0.018 [0.010]	0.019 [0.012]	0.021 [0.013]	0.021 [0.013]	0.022 [0.013]
N:	9,954	10,180	10,597	10,636	41,772	45,332	48,713	59,647

5.2 Table 2: Baseline means for wage earners

Read:

- Table 2 compares experimental and nonexperimental states before and after the state mandates and the federal PDA.
- It reports demographic variables, wages, unionization, industry mix, weekly mandate cost, and cost-to-wage ratios.
- Average hourly wages differ across experimental and nonexperimental states, which motivates using differences and fixed controls rather than raw comparisons.
- The weekly cost of the mandate averages roughly \$3–\$4 across groups, about 2 percent of wages.
- The federal PDA sample is much larger than the state-law sample because it uses a broader set of states and later CPS data.

Economic message: The table shows that the mandate cost is modest relative to wages but not negligible. It also shows why the empirical design must control for baseline differences across states and groups.

5.3 Table 3: Triple-difference wage estimate for state mandates

Read:

- The treatment group is married women aged 20–40.
- In experimental states, their mean log hourly wage falls from 1.547 to 1.513, a change of -0.034 .
- In nonexperimental states, their mean log hourly wage rises from 1.369 to 1.397, a change of $+0.028$.
- The difference-in-difference for the treatment group is therefore -0.062 .
- The control group shows a smaller difference-in-difference of -0.008 .
- The DDD estimate is -0.054 with a standard error of 0.026.

Economic message: The basic DDD estimate implies that wages of the targeted group fell by about 5.4 percent relative to controls after the state mandates. This is the paper’s simplest evidence of wage shifting.

TABLE 3—DDD ESTIMATES OF THE IMPACT OF STATE MANDATES ON HOURLY WAGES

Location/year	Before law change	After law change	Time difference for location
A. Treatment Individuals: Married Women, 20–40 Years Old:			
Experimental states	1.547 (0.012) [1,400]	1.513 (0.012) [1,496]	– 0.034 (0.017)
Nonexperimental states	1.369 (0.010) [1,480]	1.397 (0.010) [1,640]	0.028 (0.014)
Location difference at a point in time:	0.178 (0.016)	0.116 (0.015)	
Difference-in-difference:	– 0.062 (0.022)		
B. Control Group: Over 40 and Single Males 20–40:			
Experimental states	1.759 (0.007) [5,624]	1.748 (0.007) [5,407]	– 0.011 (0.010)
Nonexperimental states	1.630 (0.007) [4,959]	1.627 (0.007) [4,928]	– 0.003 (0.010)
Location difference at a point in time:	0.129 (0.010)	0.121 (0.010)	
Difference-in-difference:	– 0.008: (0.014)		
DDD:	– 0.054 (0.026)		

Notes: Cells contain mean log hourly wage for the group identified. Standard errors are given in parentheses; sample sizes are given in square brackets. Years before/after law change, and experimental/nonexperimental states, are defined in the text. Difference-in-difference-in-difference (DDD) is the difference-in-difference from the upper panel minus that in the lower panel.

TABLE 4—TREATMENT-DUMMY RESULTS ACROSS DEMOGRAPHIC GROUPS

Group	Log hourly wage	Log hours/week	Employment (probit)	Percentage changes in labor input
Married women, ages 20–40	-0.043 (0.023)	0.049 (0.022)	-0.047 (0.048) [-0.016]	1.40
Single women, ages 20–40	-0.042 (0.026)	-0.014 (0.024)	-0.095 (0.064) [-0.030]	-5.95
Married men, ages 20–40	-0.009 (0.018)	0.030 (0.015)	-0.139 (0.072) [-0.038]	-1.08
All treatments	-0.023 (0.015)	0.027 (0.014)	-0.079 (0.039) [-0.024]	-0.88

Notes: Standard errors are given in parentheses. The coefficient is that on the third-level interaction in equation (1). The treatment group is the group indicated for each row. The control group is the same as that for Table 3 (all those older than 40 and single men younger than 40). The number in brackets in the employment column is the marginal probability (see text). The change in total labor input is the change in hours at the average-employment/population ratio plus the change in employment in terms of average hours per employed person. This is then divided by the ratio of employment to population to get per-worker figures and then divided by average hours per week for the treatment group to get a percentage change.

5.4 Table 4: Treatment-dummy estimates across demographic groups

Read:

- Married women aged 20–40 have a log wage effect of -0.043 , close to the Table 3 DDD estimate.
- Single women aged 20–40 have a similar log wage effect of -0.042 .
- Married men aged 20–40 have a smaller wage effect of -0.009 .
- Combining all treatment groups gives a wage coefficient of -0.023 .

- Hours and employment estimates vary by group, but the changes in total labor input are small overall.

Economic message: The wage effects appear not only for married women but also for other groups whose insurance costs may rise. The quantity effects are modest, supporting the incidence interpretation.

TABLE 4—TREATMENT-DUMMY RESULTS ACROSS DEMOGRAPHIC GROUPS

Group	Log hourly wage	Log hours/week	Employment (probit)	Percentage changes in labor input
Married women, ages 20–40	−0.043 (0.023)	0.049 (0.022)	−0.047 (0.048) [−0.016]	1.40
Single women, ages 20–40	−0.042 (0.026)	−0.014 (0.024)	−0.095 (0.064) [−0.030]	−5.95
Married men, ages 20–40	−0.009 (0.018)	0.030 (0.015)	−0.139 (0.072) [−0.038]	−1.08
All treatments	−0.023 (0.015)	0.027 (0.014)	−0.079 (0.039) [−0.024]	−0.88

Notes: Standard errors are given in parentheses. The coefficient is that on the third-level interaction in equation (1). The treatment group is the group indicated for each row. The control group is the same as that for Table 3 (all those older than 40 and single men younger than 40). The number in brackets in the employment column is the marginal probability (see text). The change in total labor input is the change in hours at the average-employment/population ratio plus the change in employment in terms of average hours per employed person. This is then divided by the ratio of employment to population to get per-worker figures and then divided by average hours per week for the treatment group to get a percentage change.

5.5 Table 5: Parametrized cost of the mandate

Read:

- Table 5 replaces a simple treatment dummy with a predicted individual-specific mandate cost.
- The first wage specification implies 214 percent shifting when cost is normalized by hours.
- Without hours normalization, the estimate implies about 109 percent shifting.
- Restricting to full-time workers gives about 156 percent shifting.
- The employment and hours estimates do not imply a large negative effect on total labor input.

Economic message: The cost-parametrized estimates reinforce the main result: workers with higher predicted maternity-benefit costs have lower wages after the mandates, with magnitudes close to full shifting.

TABLE 5—WAGES AND LABOR INPUT RESULTS—PARAMETRIZED COST OF THE MANDATE

Coefficient	Specification				
	(i) Log wage	(ii) Log wage (no hours)	(iii) Log wage (full-time)	(iv) Log hours/week	(v) Employment (probit)
β_8	−2.140 (0.759)	−0.028 (0.019)	−0.037 (0.019)	0.0049 (0.0031)	−0.027 (0.011) [−0.022]
Shifting (percentage):	214	109	156	41,367	84,305
N:	41,367	41,367	35,868	41,367	84,305

Notes: All regressions are estimated by nonlinear least squares, as described in the text. Standard errors are given in parentheses. The mandate cost in columns (ii) and (iii) is not normalized by hours worked; shifting is calculated at average hours for the treatments. The sample in column (iii) is restricted to those who work at least 35 hours per week. Column (v) is a probit. Cost is assigned by demographic group average. The number in brackets shows the change in the probability of employment for a \$1 increase in costs.

TABLE 6—FURTHER RESULTS—FEDERAL EXPERIMENT

Demographic group/treatment	Specification				
	(i) Log wage	(ii) Log wage (no hours)	(iii) Log wage (full-time)	(iv) Log hours/week	(v) Employment (probit)
Married women, ages 20–40	−0.021 (0.012)			0.0012 (0.0098)	−0.018 (0.028) [−0.0055]
Single women, ages 20–40	−0.014 (0.014)			0.0157 (0.0101)	0.0184 (0.0374) [0.0050]
Married men, ages 20–40	−0.008 (0.0012)			−0.0008 (0.0073)	0.0020 (0.0046) [0.0005]
All treatments	−0.0014 (0.0009)			0.0032 (0.0064)	0.0001 (0.0233) [0.00004]
Individual parameterization	−0.587 (0.412)	−0.023 (0.010)	−0.017 (0.010)	−0.0002 (0.0015)	0.0007 (0.0068) [0.00005]
Shifting (percentage):	59	90	75		

Notes: The coefficient is β_8 in equation (1). Standard errors are given in parentheses. In column (v), the number in brackets interprets the probit coefficient for employment, by calculating the change in probability of employment for a \$1 increase in the cost of the mandate. Change in total labor input is the change in total hours per week per worker for a 100-percent rise in the cost of the mandate. It is calculated by adding the change in hours at average employment to the change in employment at average hours, for a \$1 rise in cost, for the relevant treatment group. This is then divided by average labor input (hours times employment/population ratio) for the treatment group (and multiplied by cost per week in the parameterized cost case) to get the percentage change in labor input for a 100-percent rise in the dollar cost.

5.6 Table 6: Federal experiment

Read:

- Table 6 repeats the analysis using the federal Pregnancy Discrimination Act.
- The first row shows wage reductions for married women aged 20–40 in newly treated states, though the demographic-dummy estimate is smaller than in the state-law experiment.
- Single women and married men show weaker wage effects.
- The individual-cost specification implies about 59 percent shifting when cost is normalized by hours, 90 percent shifting without hours normalization, and 75 percent shifting for full-time workers.
- The labor-input estimates remain statistically weak and do not show large employment or hours losses.

Economic message: The federal reverse experiment supports the same qualitative conclusion as the state mandates: maternity-benefit costs were substantially shifted to wages, with little evidence of large labor-input distortions.

TABLE 6—FURTHER RESULTS—FEDERAL EXPERIMENT

Demographic group/treatment	Specification					Change in total labor input
	(i) Log wage	(ii) Log wage (no hours)	(iii) Log wage (full-time)	(iv) Log hours/week	(v) Employment (probit)	
Married women, ages 20–40	−0.021 (0.012)			0.0012 (0.0098)	−0.018 (0.028) [−0.0055]	−0.0071
Single women, ages 20–40	−0.014 (0.014)			0.0157 (0.0101)	0.0184 (0.0374) [0.0050]	0.0219
Married men, ages 20–40	−0.008 (0.0012)			−0.0008 (0.0073)	0.0020 (0.0046) [0.0005]	−0.0003
All treatments	−0.0014 (0.0009)			0.0032 (0.0064)	0.0001 (0.0233) [0.00004]	0.0033
Individual parameterization	−0.587 (0.412)	−0.023 (0.010)	−0.017 (0.010)	−0.0002 (0.0015)	0.0007 (0.0068) [0.00005]	−0.0005
Shifting (percentage):	59	90	75			

Notes: The coefficient is β_3 in equation (1). Standard errors are given in parentheses. In column (v), the number in brackets interprets the probit coefficient for employment, by calculating the change in probability of employment for a \$1 increase in the cost of the mandate. Change in total labor input is the change in total hours per week per worker for a 100-percent rise in the cost of the mandate. It is calculated by adding the change in hours at average employment to the change in employment at average hours, for a \$1 rise in cost, for the relevant treatment group. This is then divided by average labor input (hours times employment/population ratio) for the treatment group (and multiplied by cost per week in the parameterized cost case) to get the percentage change in labor input for a 100-percent rise in the dollar cost.

6 Interpretation of the empirical findings

6.1 Incidence of the mandate

- The paper consistently finds negative wage effects for groups whose expected maternity-insurance costs rose.
- The wage reductions are of roughly the same order as the actuarial cost of adding maternity benefits.
- This indicates that workers, rather than firms, bore much of the cost through lower wages.

Interpretation: The result is analogous to tax incidence. Even though the law formally places the mandate on employers, the economic burden falls on workers when wages adjust.

6.2 Labor-input effects

- If employers could not shift costs, they might reduce employment or hours of affected workers.
- The paper generally does not find large net decreases in labor input.
- Some estimates show compositional changes, such as hours changes or employment changes for specific groups, but the net effects are small and imprecise.

Interpretation: The lack of large labor-input effects is consistent with full or near-full wage shifting. If wages absorb the cost, employer demand for the targeted group should not fall much.

6.3 Efficiency implications

- If workers value maternity benefits approximately at cost, wage shifting means the mandate may have low efficiency costs.
- The mandate functions like a benefit tax: workers receive insurance and pay through lower wages.
- Because the group-specific wage adjusts, the mandate does not necessarily distort relative hiring costs across groups.

Interpretation: The efficiency case for mandates depends on worker valuation. The paper estimates incidence, not directly the value workers place on the benefit.

6.4 Equity implications

- Wage shifting can undermine redistribution. If the goal is to transfer resources toward women of childbearing age, shifting the cost to those same women limits redistribution.
- Mandates may be regressive relative to broad tax finance if low-wage workers pay through lower wages.
- The maternity-benefit case may also raise questions about overinsurance and medical utilization.

Interpretation: The paper's results support the efficiency possibility of mandates but do not imply that mandates are always desirable on equity grounds.

7 Short summary

- Gruber studies state and federal mandates requiring comprehensive maternity coverage in employer health insurance.
- Before the reforms, maternity benefits were often excluded or less generous than coverage for comparable illnesses.
- The mandates raised the cost of insuring women of childbearing age and, through family coverage, some men.
- The paper uses CPS data and quasi-experimental variation across states, demographic groups, and time.
- The state-law DDD estimate shows a wage decline of about 5.4 percent for married women aged 20–40.
- Treatment-dummy regressions show negative wage effects for married women, single women, and the combined affected group.
- Individual cost-parametrized estimates imply roughly full shifting of mandate costs to wages.
- The federal PDA experiment confirms the qualitative finding of substantial wage shifting.
- The paper finds little evidence of a large decline in employment, hours, or total labor input.
- The main conclusion is that the economic cost of mandated maternity benefits was largely borne by affected workers through lower wages.

8 Conclusion

8.1 Core conclusion

Gruber finds that mandated maternity benefits were substantially shifted to the wages of the targeted workers. Although employers were legally required to provide more generous insurance coverage, wages adjusted

downward by an amount close to the actuarial cost of the mandate.

8.2 Economic intuition

The mandate raises the cost of employing a group. If members of the group value the benefit and if relative wages can adjust, the cost can be passed back to workers through lower wages. In that case, the mandate does not substantially change employers' net cost of hiring the group, so employment and hours need not fall much.

8.3 Takeaway

The paper shows that the statutory incidence of a benefit mandate differs from its economic incidence. Mandates can finance benefits through worker wages, even for group-specific mandates. This supports the view that mandates may be an efficient financing device, but it also means they may fail to redistribute resources toward the intended beneficiaries.